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Cloud Infrastructure and Services

25CS2206E

LAB RECORD

2026

EXPERIMENT – 2

Title: Enabling Multi-Factor Authentication, Creating a Linux EC2 Instance, and S3 Bucket Creation with Object Upload

AIM

To improve AWS account security by using multi-factor authentication (MFA) or a passkey-based additional verification method, launch a Linux virtual machine using Amazon EC2, and create an Amazon S3 bucket to store and upload objects with different file extensions.

DESCRIPTION

AWS provides identity security through additional authentication factors, compute capacity through Amazon EC2 instances, and scalable object storage through Amazon S3. MFA adds an additional verification step beyond a password. EC2 allows users to launch and connect to cloud-based Linux virtual machines. S3 stores files as objects inside buckets. This practical covers the sign-in verification flow, launching an Amazon Linux instance, connecting to it, and creating an S3 bucket.

REQUIREMENTS

- A valid AWS account and access to the AWS Management Console.
- An authenticator/passkey or security-key method supported by the account and device.
- An EC2 key pair and a Linux AMI (Amazon Linux 2023 is shown in the screenshots).
- Internet access and a browser; an SSH/EC2 Instance Connect session for Linux commands.
- Five sample files with different extensions for the S3 upload activity.

A) ENABLING MULTI-FACTOR AUTHENTICATION TO SECURE YOUR ACCESS

Procedure:

1. Open the AWS Management Console sign-in page.

2. Select the appropriate account type. The supplied screenshots show the Root user option and the account email address.

3. Enter the account password and continue.

4. Complete the additional verification prompt. The supplied evidence shows a Windows Security passkey/security-key prompt asking the user to verify identity using the device fingerprint reader.

5. After successful verification, confirm that the AWS Management Console opens. For account security, MFA/passkey enrollment and recovery options should be reviewed in the account's security settings.

Note: The supplied screenshots document a passkey-based additional sign-in verification step. They do not show the full IAM or root-account MFA device enrollment/settings page. If the instructor requires proof of an MFA device being enrolled, capture the AWS Security credentials or IAM Security credentials page showing the MFA device status.

B) CREATING THE FIRST LINUX INSTANCE

Procedure:

1. Sign in to the AWS Management Console and open the Services menu.

2. Search for and open EC2. Check that the selected AWS Region is the required region; the screenshots show Europe (Stockholm), eu-north-1.

3. From the EC2 Dashboard, select Launch instance.

4. Enter an instance name. The supplied screenshots use the name JARVIS.

5. Choose an Amazon Machine Image (AMI). The screenshots show Amazon Linux 2023, marked Free tier eligible.

6. Choose the instance type available for the lab account and review the free-tier eligibility and estimated charges.

7. Under Key pair (login), select an existing key pair. The screenshot shows the key pair named JARVIS. Ensure that the private key file is safely stored.

8. Review Network settings and configure access. Allow SSH (TCP port 22) only from your own IP address where possible; avoid unrestricted access from 0.0.0.0/0.

9. Review Configure storage. The screenshots show the root volume configuration, then proceed to Launch instance.

10. Wait for the success message and open the instance details page. Confirm that the instance reaches the Running state and passes its status checks.

11. Connect using EC2 Instance Connect or SSH. The supplied terminal screenshot shows an Amazon Linux 2023 shell prompt.

12. Run basic Linux commands such as pwd, ls, and mkdir JARVIS to verify terminal access and create a working directory.

C) S3 BUCKET CREATION AND UPLOAD OBJECTS

I. Create an S3 bucket for file storage

1. Open Amazon S3 from the AWS Management Console.

2. Select Buckets and choose Create bucket.

3. Enter a globally unique bucket name. The supplied screenshots show the bucket name anudeepptnuru.

4. Choose the required AWS Region. The screenshots show Europe (Stockholm), eu-north-1.

5. Keep Block Public Access enabled unless the lab specifically requires public access. Do not make the bucket public for a basic file-storage exercise.

6. Review the remaining settings and select Create bucket.

7. Open the newly created bucket and confirm that it appears in the bucket list.

II. Upload five objects with different file extensions

Procedure:

1. Open the created bucket and select Upload.

2. Choose Add files and select five files from the computer, each with a different extension.

3. Review the selected files and start the upload.

4. Wait for the Upload succeeded message and verify that all five objects appear in the Objects tab.

Suggested file types for the five objects: .txt, .pdf, .jpg, .png, and .csv. These are examples; use five files that are available on your computer and ensure that their extensions are different.

SCREENSHOT EVIDENCE – MFA / SIGN-IN VERIFICATION

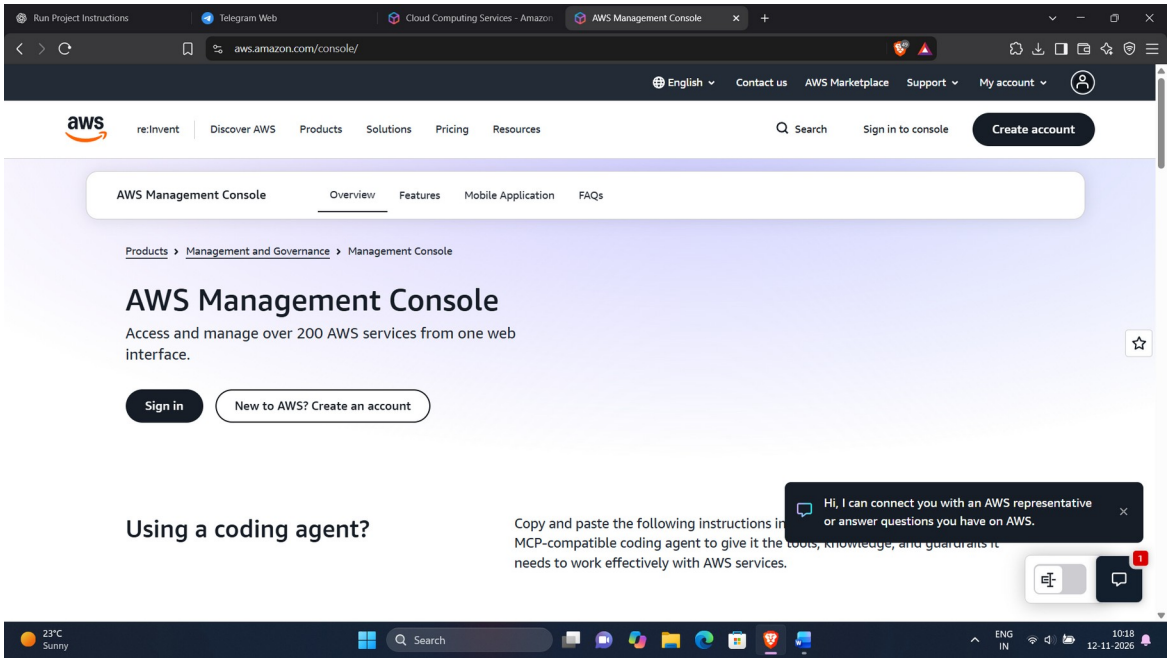


Figure 1. AWS Management Console landing page before sign-in.

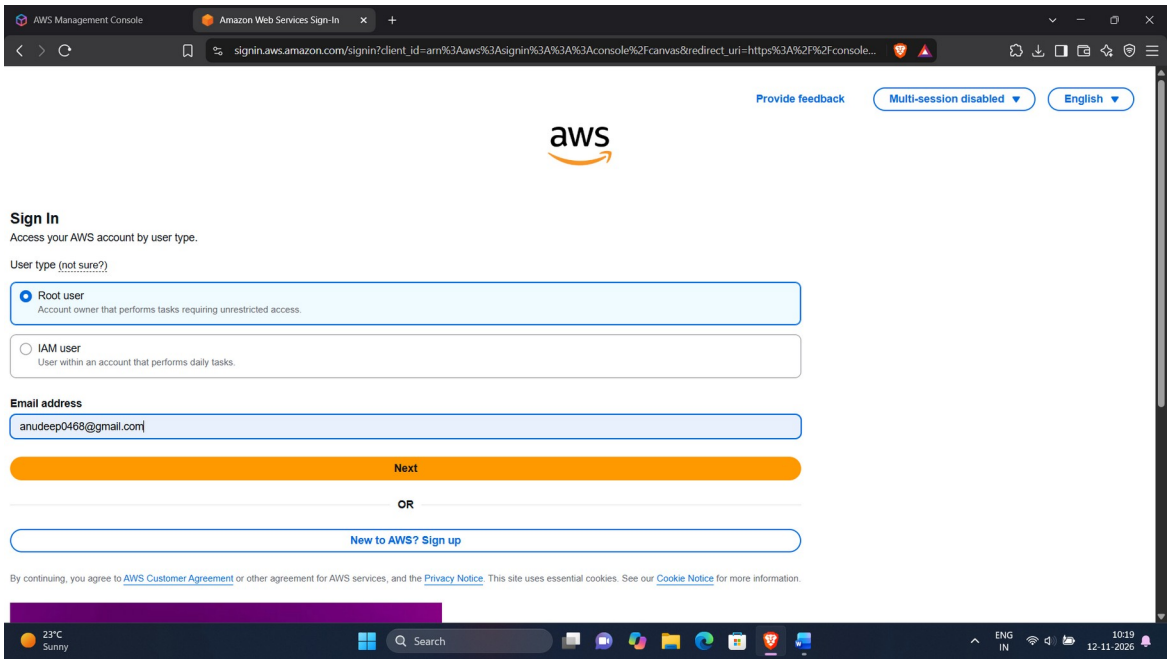


Figure 2. AWS sign-in page with Root user selected.

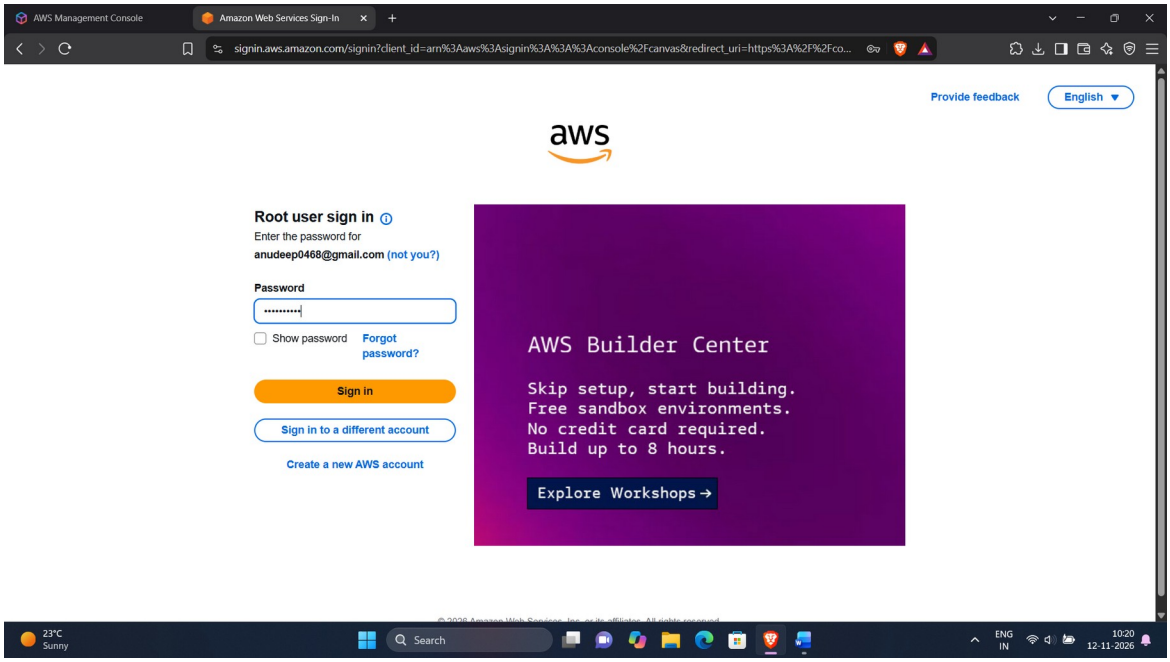


Figure 3. Root user password sign-in page.

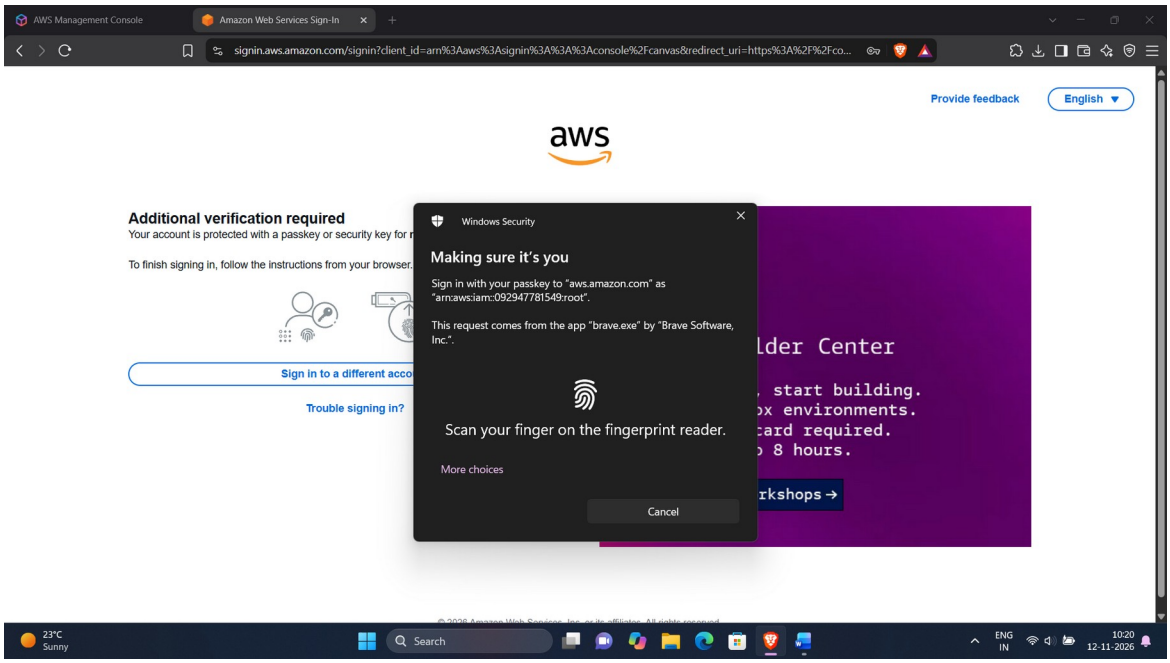


Figure 4. Additional verification prompt using Windows Security/passkey and fingerprint reader.

SCREENSHOT EVIDENCE – EC2 LINUX INSTANCE

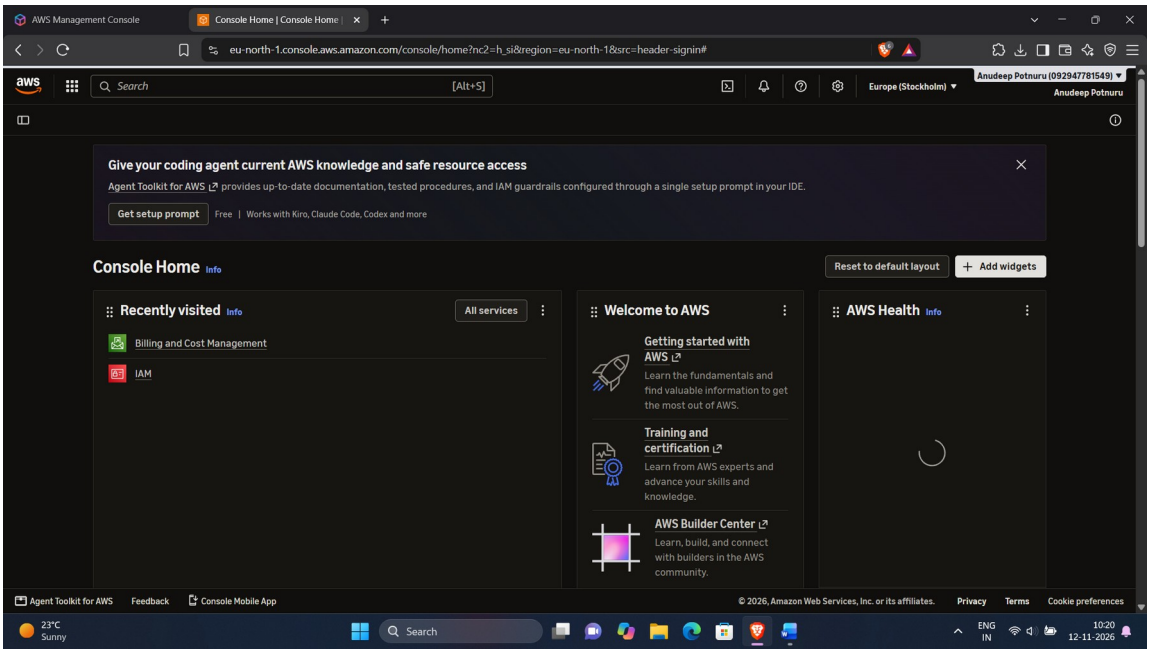


Figure 5. AWS Management Console home page after sign-in.

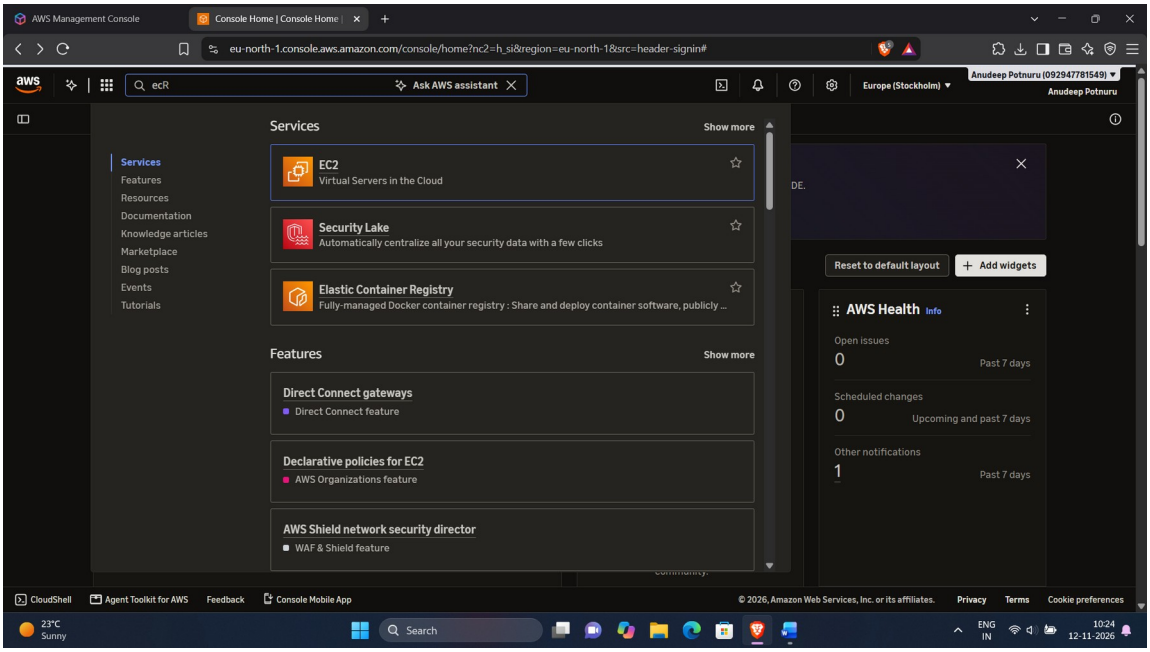


Figure 6. AWS Services menu with EC2 selected.

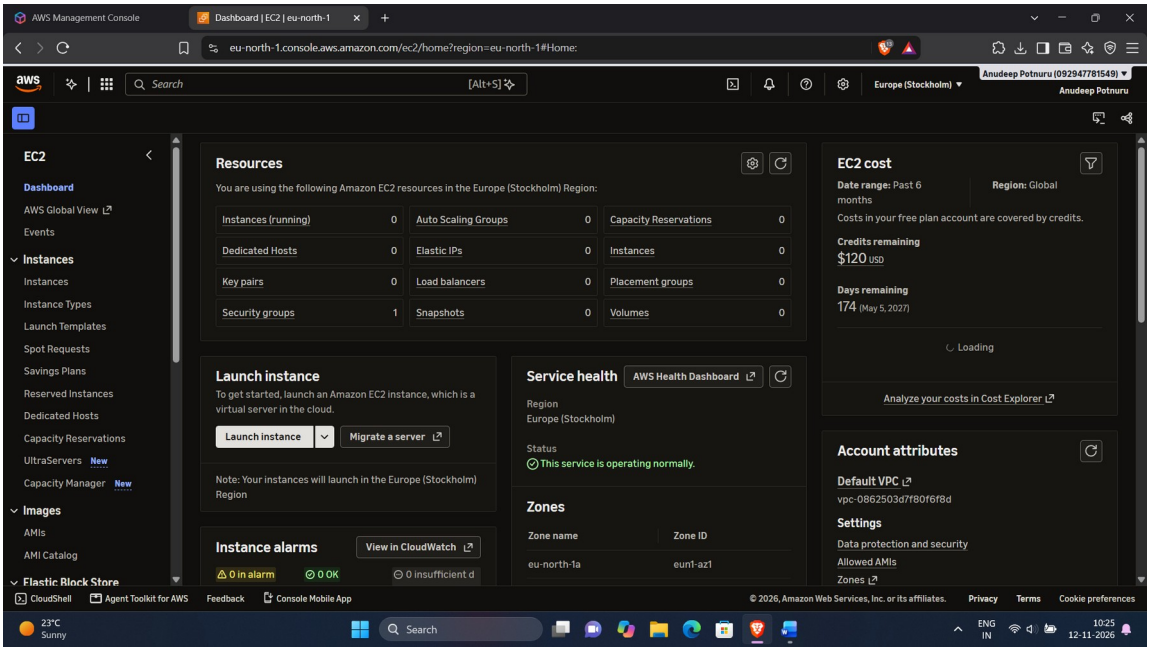


Figure 7. EC2 Dashboard in the Europe (Stockholm) Region.

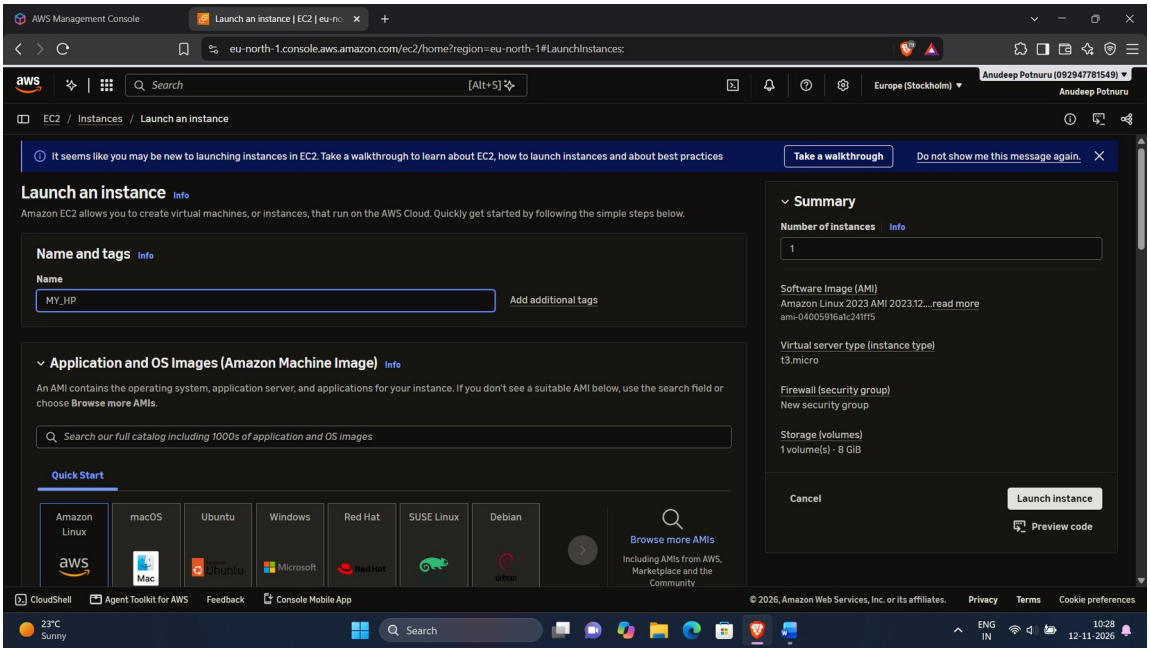


Figure 8. Launch an instance page with instance name JARVIS.

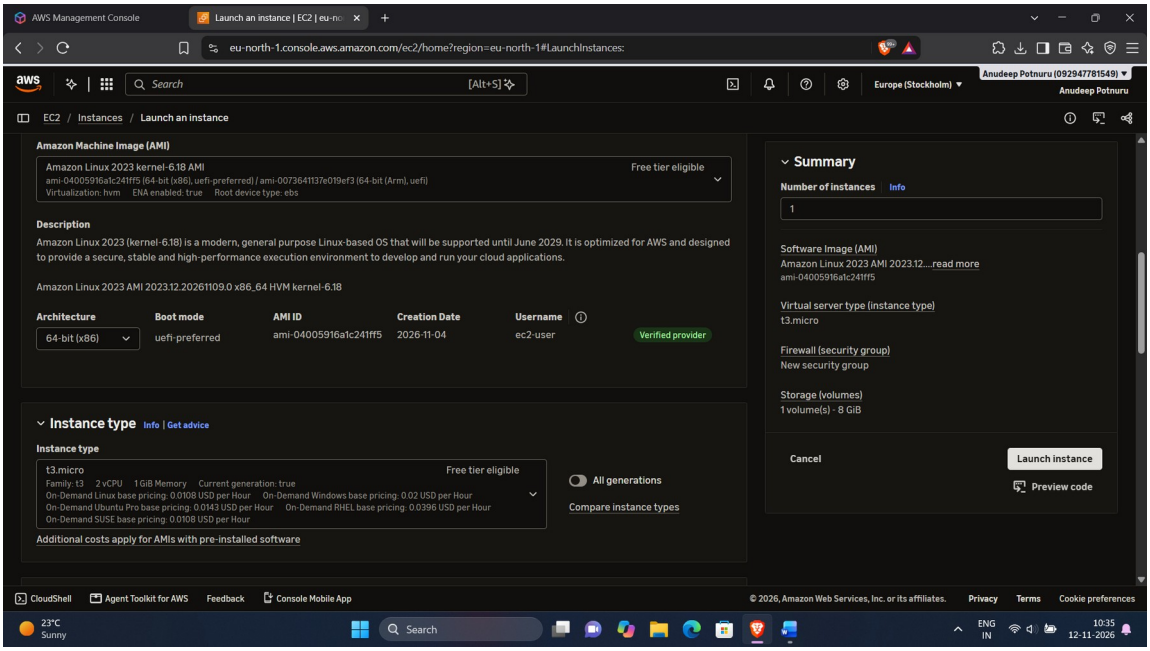


Figure 9. Amazon Linux 2023 AMI selected.

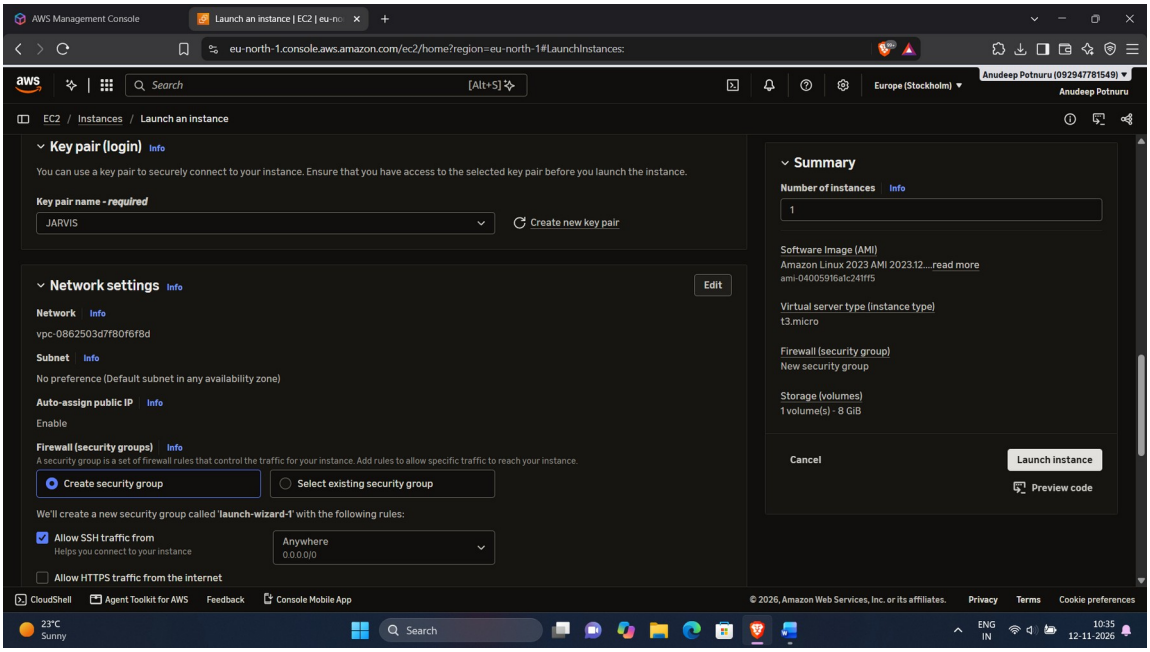


Figure 10. Key pair (login) selection; JARVIS key pair shown.

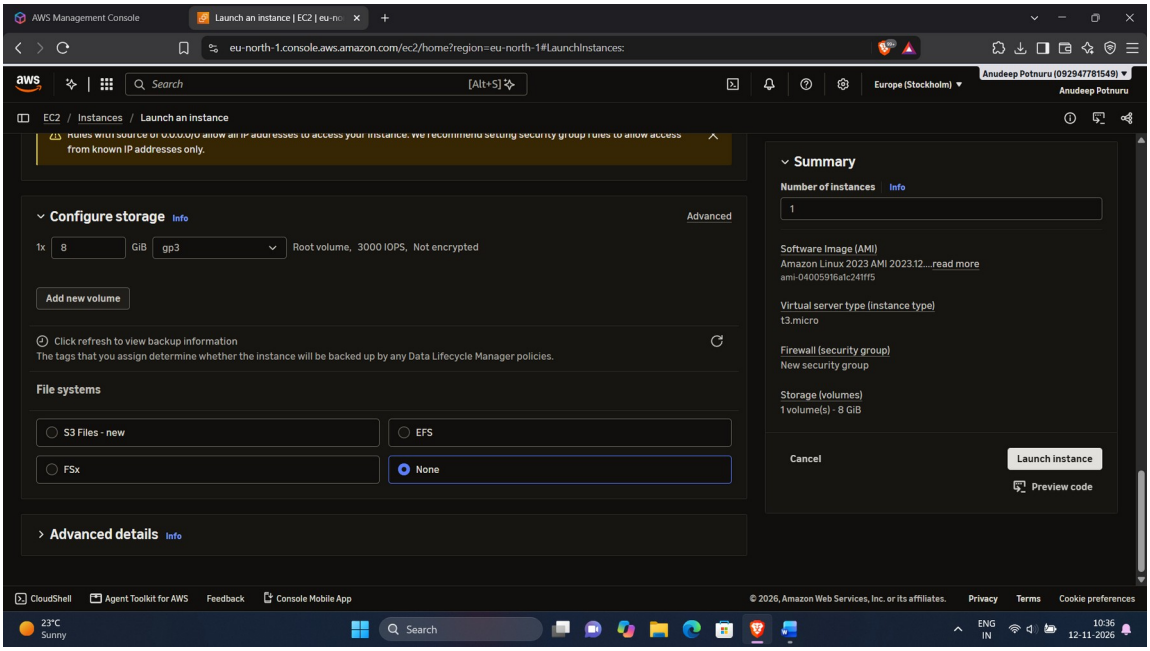


Figure 11. EC2 storage configuration review.

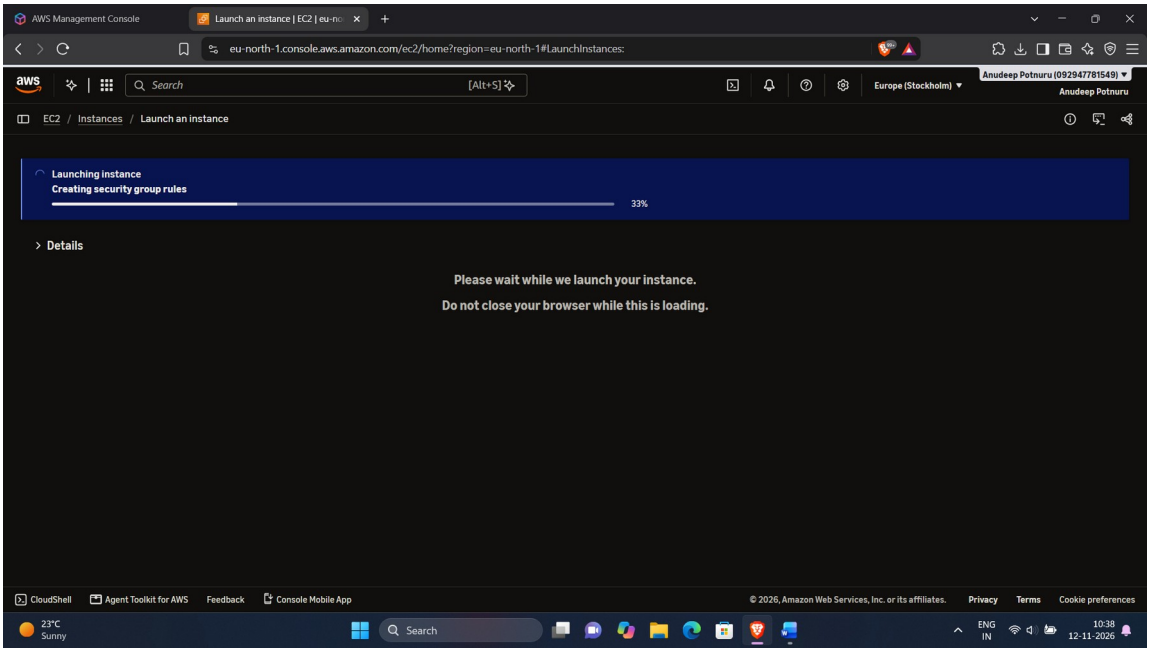


Figure 12. Instance launch in progress.

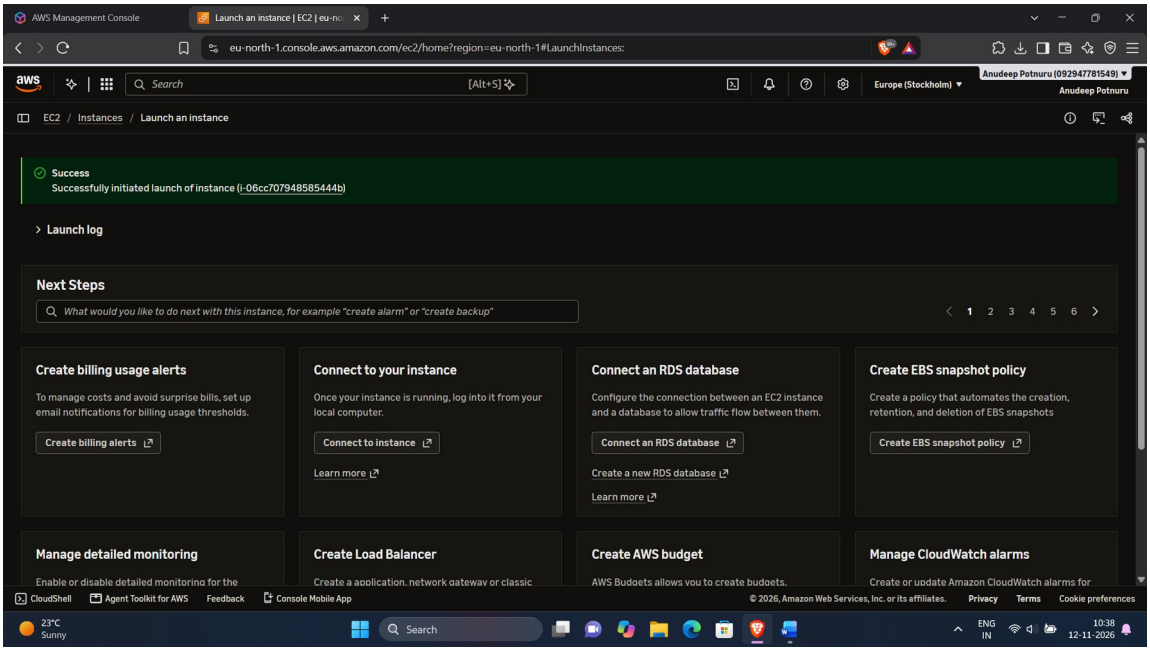


Figure 13. Successful instance launch confirmation.

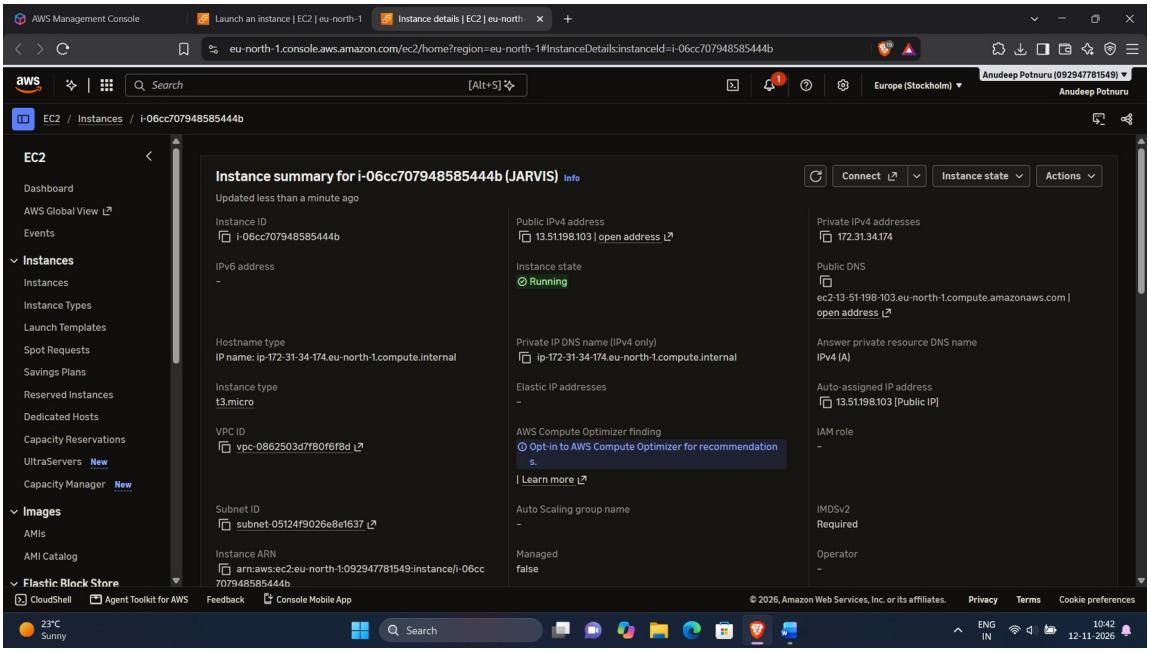


Figure 14. EC2 instance summary page.

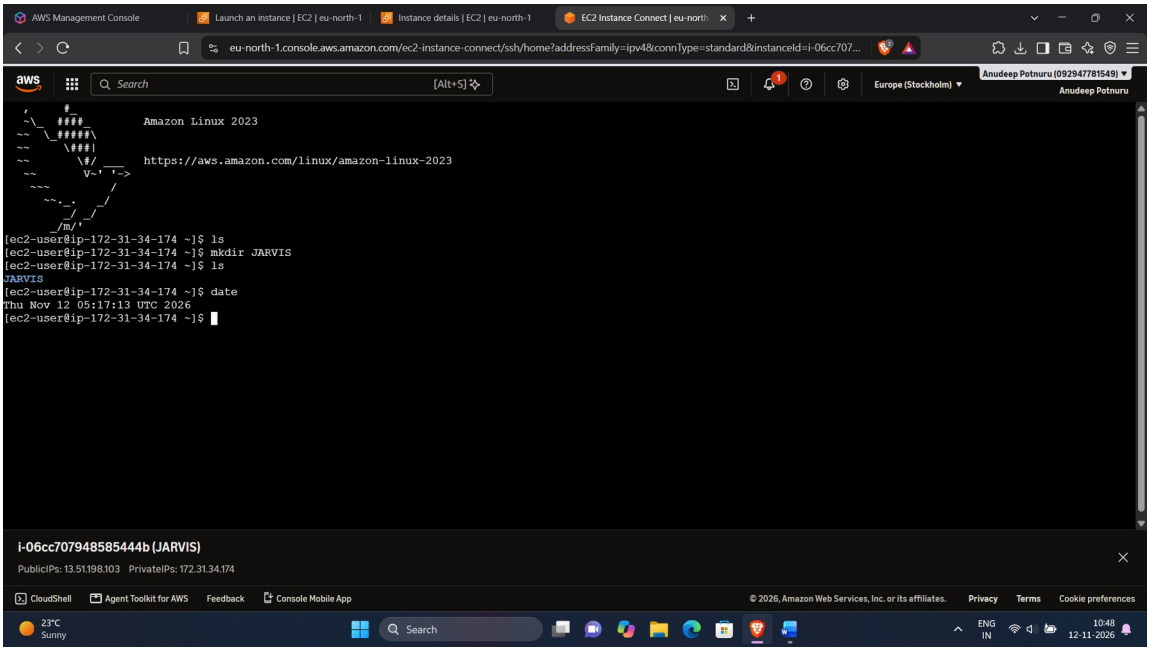


Figure 15. EC2 Instance Connect terminal on Amazon Linux 2023; basic Linux commands and JARVIS directory are shown.

SCREENSHOT EVIDENCE – AMAZON S3 BUCKET

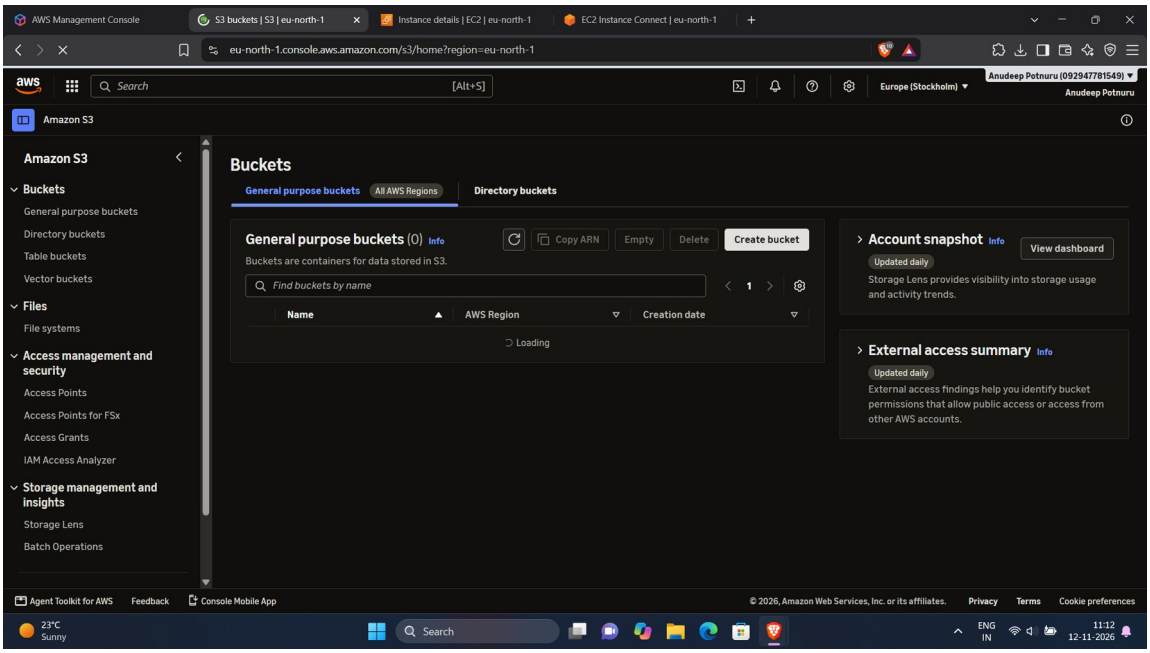


Figure 16. Amazon S3 Buckets page before bucket creation.

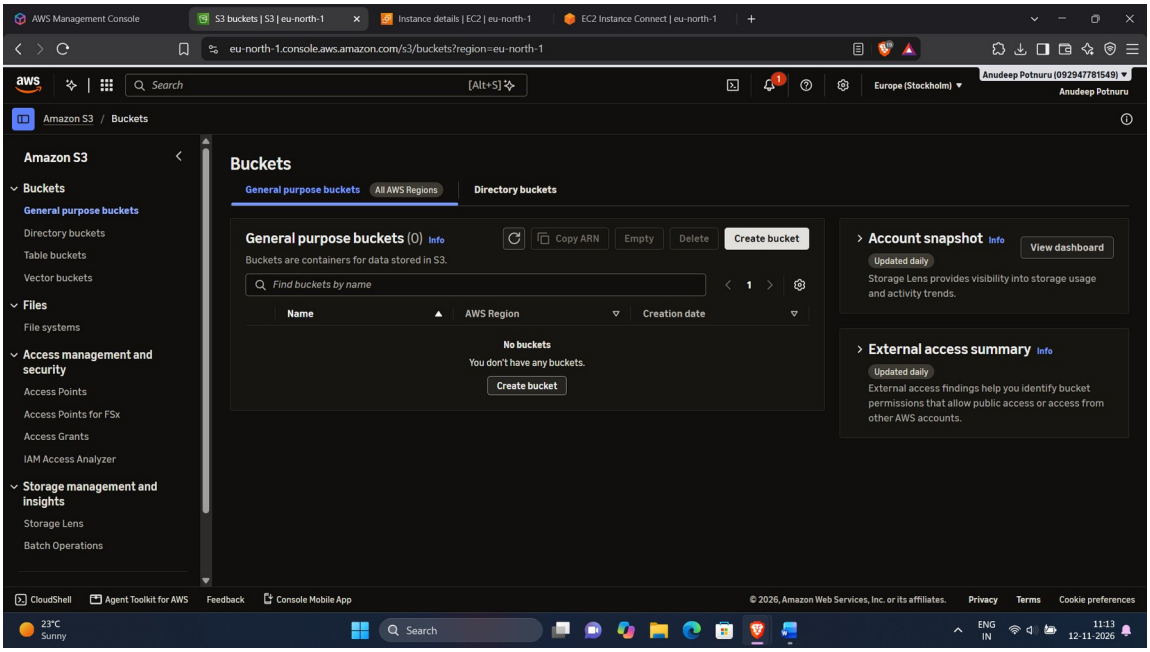


Figure 17. S3 bucket list view.

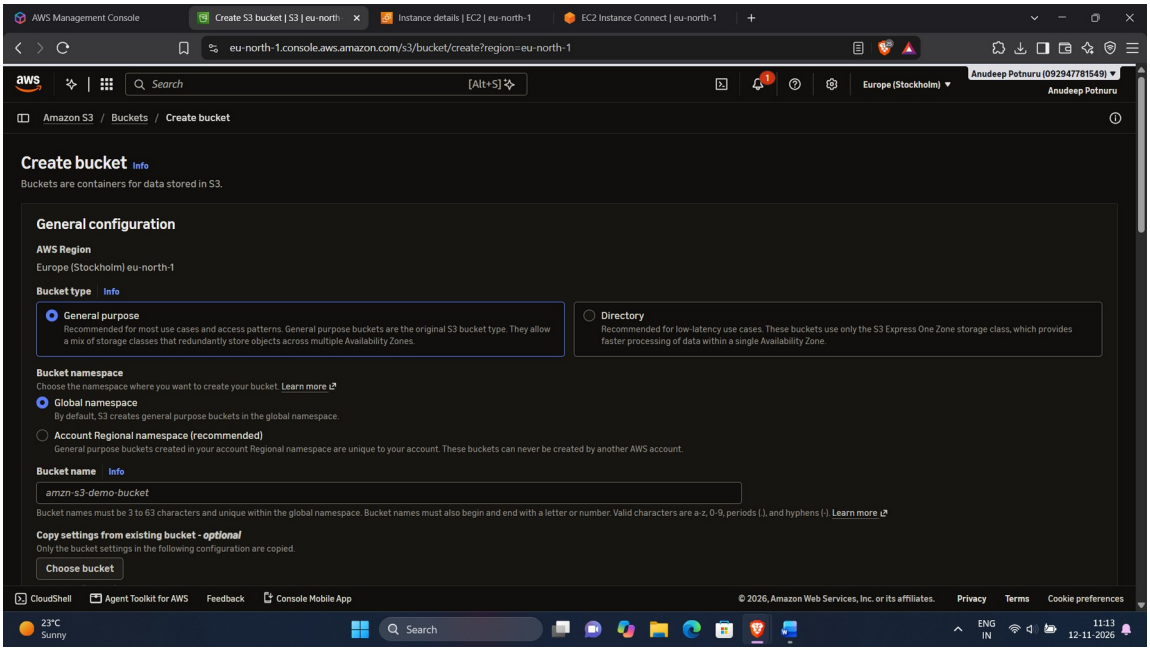


Figure 18. Create bucket page with general-purpose bucket settings.

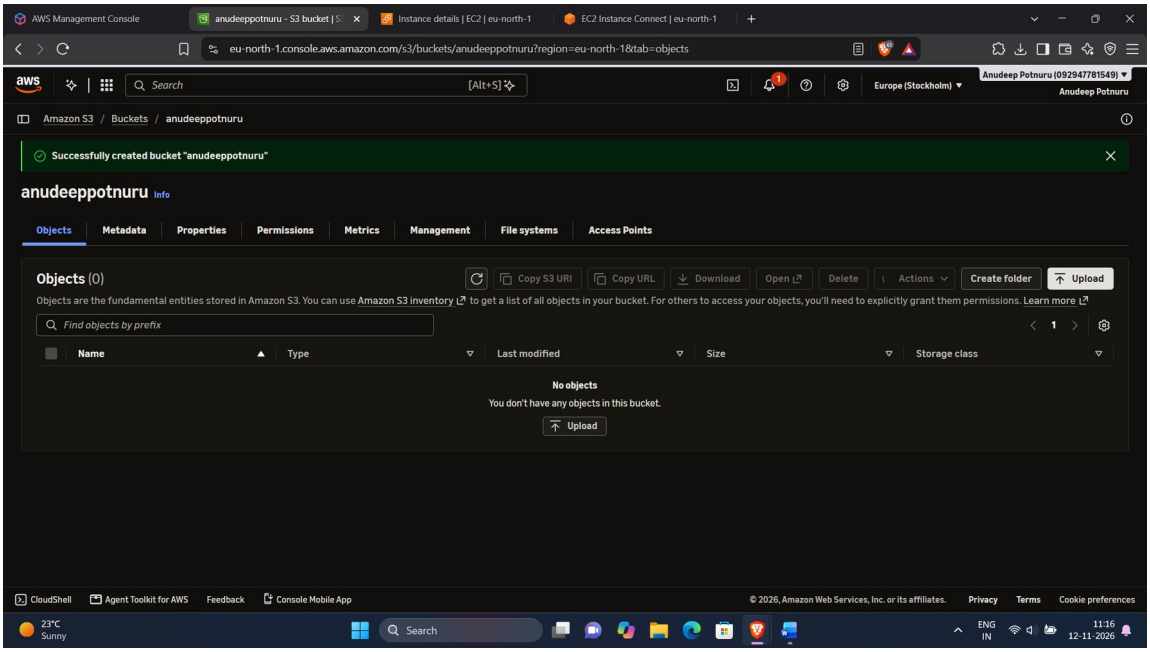


Figure 19. Successful bucket creation; bucket name anudeepptnuru is visible.

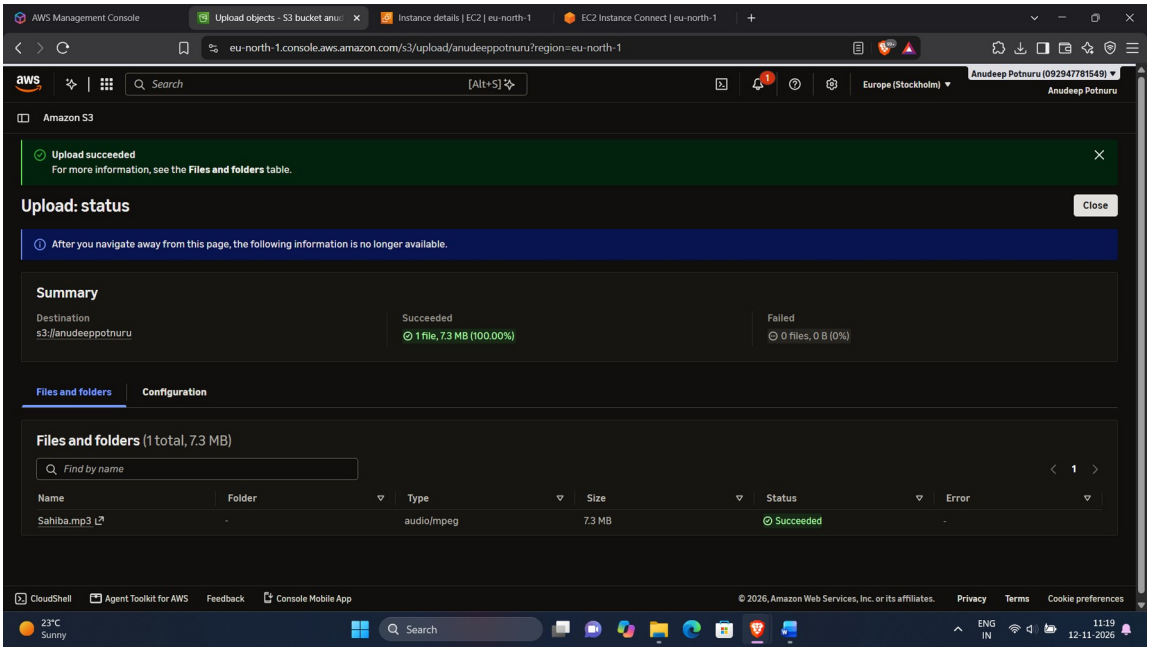


Figure 20. S3 upload status showing one file successfully uploaded to the bucket.

OUTPUT / RESULT

The supplied practical screenshots show AWS sign-in with an additional passkey/fingerprint verification prompt, access to the AWS Management Console, creation and successful launch of an Amazon Linux 2023 EC2 instance named JARVIS, connection to the Linux terminal, creation of the S3 bucket anudeepptnuru, and a successful S3 upload. The upload-status screenshot (Figure 20) reports one file uploaded successfully, not five. Therefore, upload four additional files with different extensions and capture the Objects tab showing all five objects before marking the five-object requirement as fully completed.

CONCLUSION

This practical demonstrates important AWS cloud activities: strengthening account sign-in with an additional verification method, provisioning and connecting to a Linux EC2 instance, and creating an S3 bucket for object storage. Security settings, key-pair protection, restricted network access, and verification of all uploaded objects are important parts of completing the exercise safely.